

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Aim	Implement adder circuits such as half/full adder on the FPGA development board.	
Exp No. 3	Date:	Enrollment No: 92301733013

Experiment 3

Implementation of Arithmetic Circuits Using FPGA

Objective

The objective of this experiment is to design and implement fundamental arithmetic circuits on an FPGA. These circuits include half adder, full adder, half subtractor, full subtractor, and a comparator. Each circuit will be designed using Verilog HDL and implemented on the FPGA using the specified pinout.

1. Half Adder

1.1. Theory

The half adder is a basic digital circuit used to add two single-bit binary numbers. It produces two outputs:

- Sum (S): The result of the XOR operation between the two inputs $\text{Sum} = A \oplus B$
- Carry (C): The result of the AND operation between the two inputs $\text{Carry} = A \text{ AND } B$

1.2. Structural Verilog Code

```

module Half_Adder(
    input A,
    input B,
    output Sum,
    output Carry
);

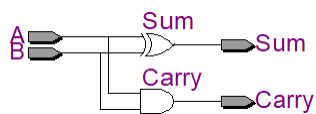
    assign Sum = A ^ B;
    assign Carry = A & B;

endmodule

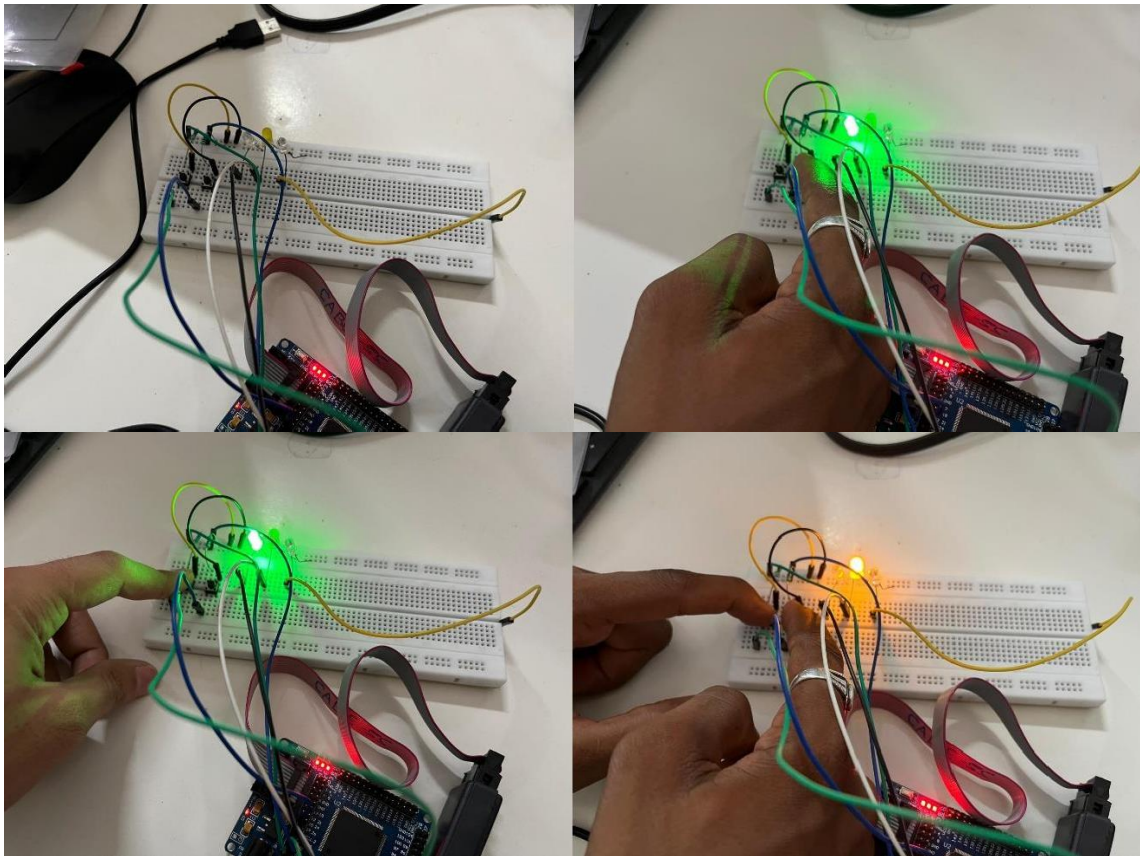
```

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1.4 RTL Schematic:



1.5 Output



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2. Full Adder

2.1. Theory

The full adder circuit adds three single-bit binary numbers, including a carry-in bit. It provides two outputs:

- Sum (S): Computed using $SUM = (A \oplus B) \oplus Cin = (A \oplus B) \oplus Cin$
- Carry Out (Cout): Computed using $CARRY-OUT = A \text{ AND } B \text{ OR } Cin(A \oplus B) = A.B + Cin(A \oplus B)$.

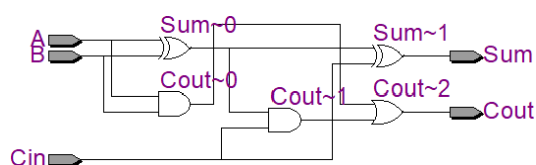
2.2. Structural Verilog Code

```
//-----
// Full Adder
//-----
module Full_Adder(
    input A,
    input B,
    input Cin,
    output Sum,
    output Cout
);

    assign Sum = A ^ B ^ Cin;
    assign Cout = (A & B) | (Cin & (A ^ B));

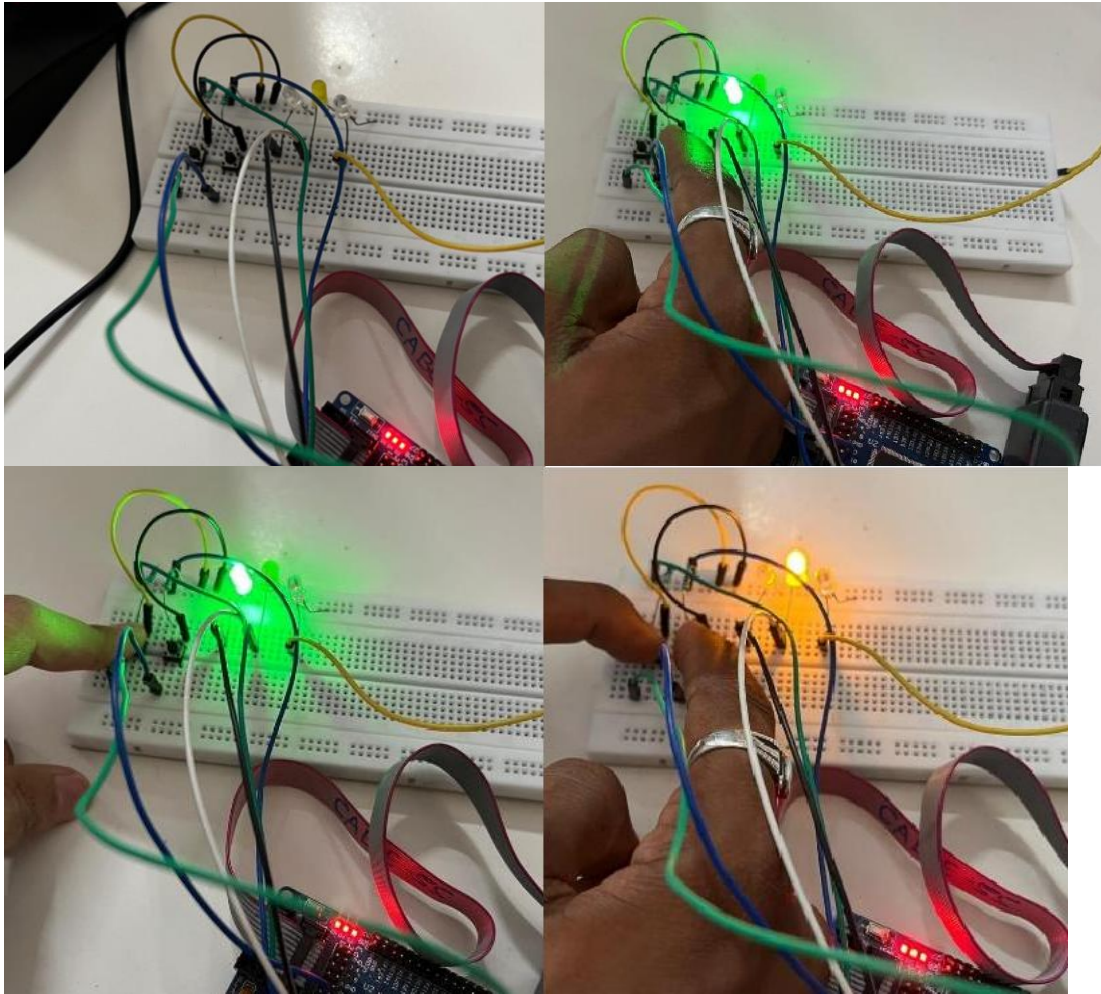
endmodule
```

2.4 RTL Schematic:



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2.5 Output



Conclusion

In this experiment, we designed and implemented **Half Adder** and **Full Adder** circuits using Verilog HDL on the FPGA development board. The designs were compiled and synthesized using **Quartus II**, programmed onto the FPGA, and then tested using the actual hardware. The obtained outputs were compared with the expected truth tables and were found to be correct. This experiment helped us understand the practical implementation of binary arithmetic circuits and gave us hands-on experience with **Verilog programming, FPGA implementation, pin assignment, synthesis, and hardware testing**. It also provided a basic foundation for developing more advanced digital circuits and systems.

Post Lab Exercise

1. What is the function of a half adder?

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A **half adder** is a combinational logic circuit used to add two single-bit binary inputs. It produces two outputs: **Sum (S)** and **Carry (C)**.

2. Which logic gate is primarily used to generate the Sum output?

The **XOR (Exclusive-OR)** gate is used to generate the Sum output of a half adder.

$$\text{Sum} = A \oplus B$$

3. What is the main difference between a half adder and a full adder?

A **Half Adder** adds two input bits, A and B, and does not have a carry input. A **Full Adder** adds three input bits: A, B, and **Carry-in (Cin)**. It produces a Sum and a **Carry-out (Cout)**.

4. Why is a full adder preferred in multi-bit addition?

A full adder is useful for multi-bit addition because it can accept a **Carry-in** from the previous bit position. Multiple full adders can therefore be connected together, allowing them to perform addition of binary numbers with multiple bits.

5. What is the purpose of implementing adders on an FPGA?

Implementing adders on an FPGA allows designers to test and verify arithmetic circuits on real hardware. It also provides a way to build high-speed digital systems such as **ALUs, processors, and digital signal-processing circuits**.

6. Which FPGA resources are commonly used to display adder outputs?

Adder outputs can be displayed using the **LEDs** and **seven-segment displays** available on an FPGA development board. The outputs can also be accessed through the board's **GPIO pins** for connection to external devices.

7. What is the advantage of FPGA implementation over software simulation?

FPGA implementation allows the circuit to run on **actual hardware in real time**. It makes it possible to test the design using physical inputs and outputs, such as switches and LEDs. Software simulation, on the other hand, tests the circuit virtually without interacting with the actual hardware.